Area of Parallelograms

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

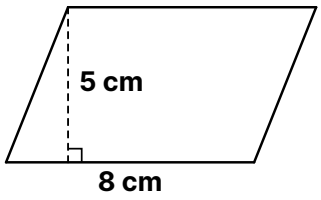
KEY WORDS parallelogram *(tapawhā whakarara)* · base *(pūtake)* · perpendicular height *(teitei hāngai)*

A Base and perpendicular height ●○○

Remember: Area = base × perpendicular height. The height is the distance between the two parallel sides, at right angles to them.

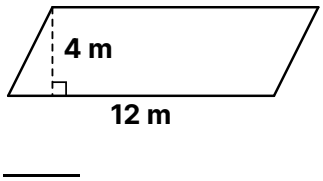
1. Find the area of each parallelogram.

a)



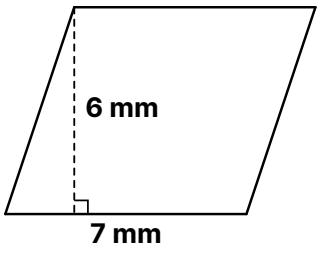
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b)



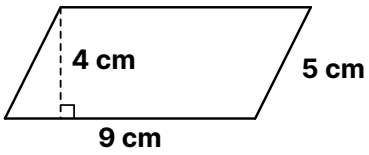
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c)



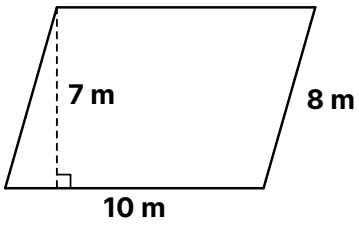
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d)



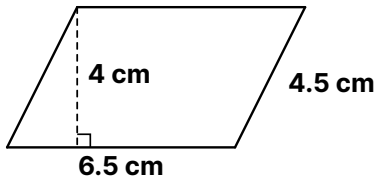
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e)



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f)



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B Finding a missing length ●●○

Remember: Put what you know into Area = base × height and solve.

2. Find the missing length.

a) A parallelogram has an area of 48 cm² and a base of 8 cm. Find its perpendicular height.

b) A parallelogram has an area of 63 m² and a perpendicular height of 7 m. Find its base.

c) A parallelogram has an area of 90 cm² and a base of 12 cm. Find its perpendicular height.

d) A parallelogram has an area of 76 mm² and a perpendicular height of 8 mm. Find its base.

e) A parallelogram has an area of 42 m² and a base of 6 m. Find its perpendicular height.

f) A parallelogram has an area of 117 cm² and a perpendicular height of 9 cm. Find its base.

g) A parallelogram has an area of 105 cm² and a base of 15 cm. Find its perpendicular height.

h) A parallelogram has an area of 96 m² and a perpendicular height of 6 m. Find its base.

i) A parallelogram has an area of 96 mm² and a base of 8 mm. Find its perpendicular height.

j) A parallelogram has an area of 130 cm² and a perpendicular height of 10 cm. Find its base.

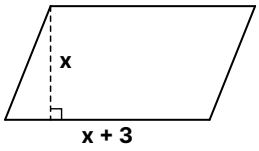
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C Lengths with algebra ●●●

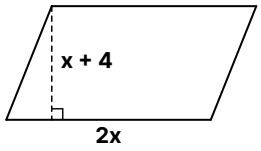
Remember: Write base × height with the expressions, then expand. To find x from a given area, write the equation and solve it.

3. Write an expression for the area of each parallelogram. Expand and simplify it.

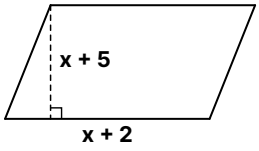
a)



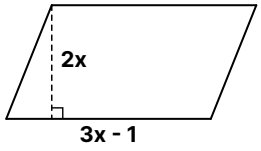
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c)

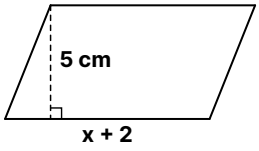


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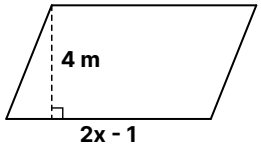
4. The area of each parallelogram is given. Write an equation and solve it to find x.

a)



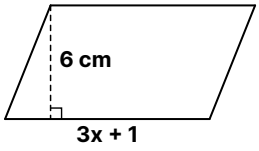
The area is 40 cm². Find x.

b)



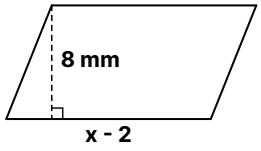
The area is 36 m². Find x.

c)



The area is 78 cm². Find x.

d)



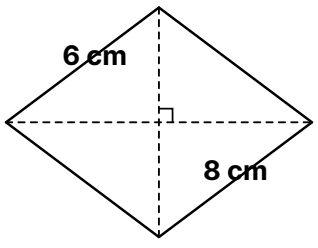
The area is 56 mm². Find x.

D Rhombuses and kites ●●●

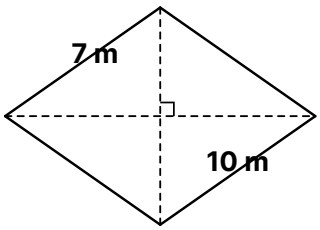
Remember: A rhombus or a kite can be cut by its diagonals into triangles. Area = $\frac{1}{2}$ × one diagonal × the other diagonal.

5. Find the area of each shape from its diagonals.

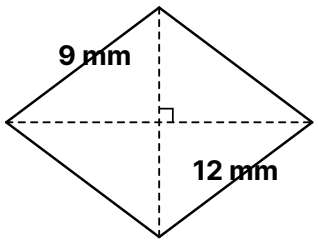
a)



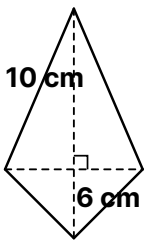
b)



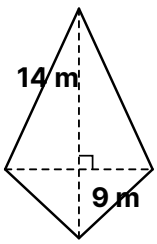
c)



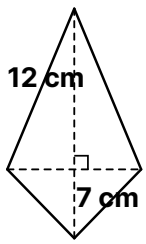
d)



e)



f)



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E Problems ●●●

6. A road sign is a parallelogram with a base of 2.4 m and a perpendicular height of 1.5 m. Reflective paint costs $38 per square metre. Find the cost of painting the sign.

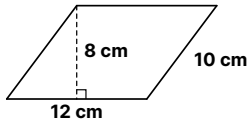
7. A parallelogram-shaped tile has a base of 30 cm and a height of 20 cm. Find the area of 240 tiles, in square metres.

8. A paddock is a parallelogram with a base of 85 m and a perpendicular height of 60 m. Fertiliser is spread at 1,250 kg per hectare (1 hectare = 10 000 m²). How much fertiliser is needed?

9. A logo is a parallelogram with a base of 6 cm and a height of 4 cm. It is enlarged by a scale factor of 2.5. Find the area of the enlarged logo.

★ Extension

10. A parallelogram has a base of 12 cm and slant sides of 10 cm. Its perpendicular height on the base is 8 cm.



a) Find its area.

b) The area is also (slant side) × (the perpendicular distance between the slant sides). Use this to find that distance.

c) Explain why a parallelogram with base 12 cm and slant side 10 cm cannot have an area of more than 120 cm².

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1-1 PRIVATE TUTORS

MEASUREMENT

Area of Parallelograms

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

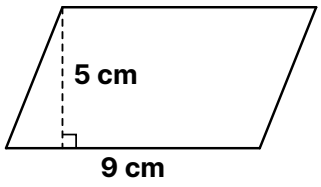
KEY WORDS parallelogram *(tapawhā whakarara)* · base *(pūtake)* · perpendicular height *(teitei hāngai)*

A Base and perpendicular height ●○○

Remember: Area = base × perpendicular height. The height is the distance between the two parallel sides, at right angles to them.

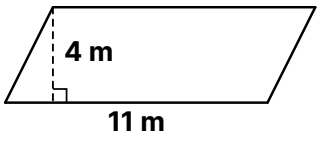
1. Find the area of each parallelogram.

a)



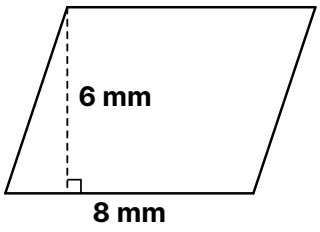
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b)



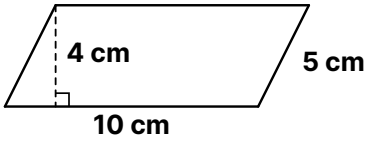
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c)



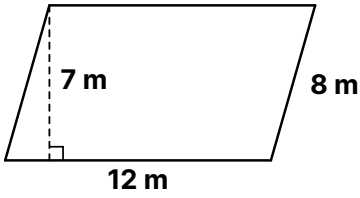
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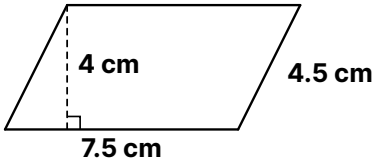
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e)



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B Finding a missing length ●●○

Remember: Put what you know into Area = base × height and solve.

2. Find the missing length.

a) A parallelogram has an area of 54 cm² and a base of 9 cm. Find its perpendicular height.

b) A parallelogram has an area of 56 m² and a perpendicular height of 7 m. Find its base.

c) A parallelogram has an area of 84 cm² and a base of 12 cm. Find its perpendicular height.

d) A parallelogram has an area of 72 mm² and a perpendicular height of 8 mm. Find its base.

e) A parallelogram has an area of 45 m² and a base of 5 m. Find its perpendicular height.

f) A parallelogram has an area of 126 cm² and a perpendicular height of 9 cm. Find its base.

g) A parallelogram has an area of 120 cm² and a base of 15 cm. Find its perpendicular height.

h) A parallelogram has an area of 84 m² and a perpendicular height of 6 m. Find its base.

i) A parallelogram has an area of 72 mm² and a base of 8 mm. Find its perpendicular height.

j) A parallelogram has an area of 140 cm² and a perpendicular height of 10 cm. Find its base.

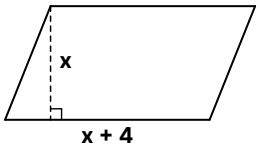
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C Lengths with algebra ●●●

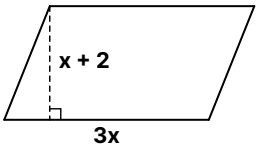
Remember: Write base × height with the expressions, then expand. To find x from a given area, write the equation and solve it.

3. Write an expression for the area of each parallelogram. Expand and simplify it.

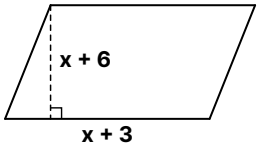
a)



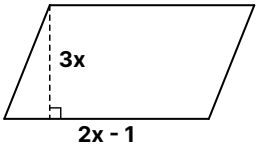
b)



c)

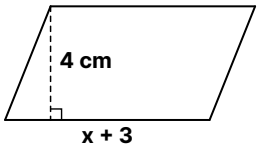


d)



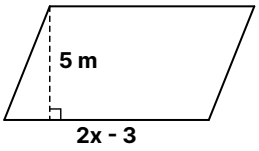
4. The area of each parallelogram is given. Write an equation and solve it to find x.

a)



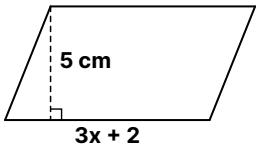
The area is 40 cm². Find x.

b)



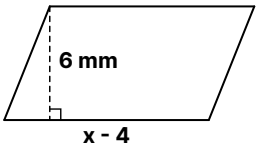
The area is 45 m². Find x.

c)



The area is 70 cm². Find x.

d)



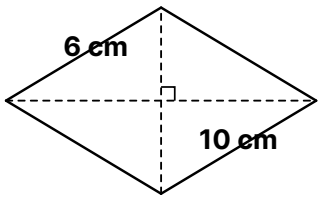
The area is 36 mm². Find x.

D Rhombuses and kites ●●●

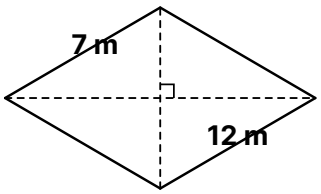
Remember: A rhombus or a kite can be cut by its diagonals into triangles. Area = $\frac{1}{2}$ × one diagonal × the other diagonal.

5. Find the area of each shape from its diagonals.

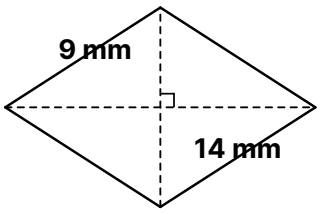
a)



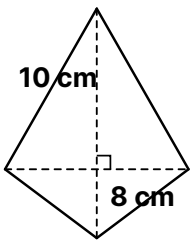
b)



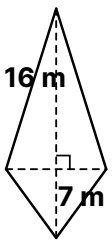
c)



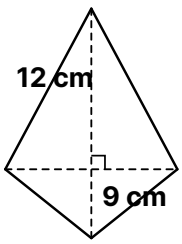
d)



e)



f)



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E Problems ●●●

6. A road sign is a parallelogram with a base of 3 m and a perpendicular height of 1.2 m. Reflective paint costs $42 per square metre. Find the cost of painting the sign.

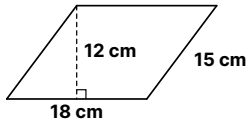
7. A parallelogram-shaped tile has a base of 25 cm and a height of 20 cm. Find the area of 300 tiles, in square metres.

8. A paddock is a parallelogram with a base of 90 m and a perpendicular height of 50 m. Fertiliser is spread at 1,200 kg per hectare (1 hectare = 10 000 m²). How much fertiliser is needed?

9. A logo is a parallelogram with a base of 8 cm and a height of 5 cm. It is enlarged by a scale factor of 3. Find the area of the enlarged logo.

★ Extension

10. A parallelogram has a base of 18 cm and slant sides of 15 cm. Its perpendicular height on the base is 12 cm.



a) Find its area.

b) The area is also (slant side) × (the perpendicular distance between the slant sides). Use this to find that distance.

c) Explain why a parallelogram with base 18 cm and slant side 15 cm cannot have an area of more than 270 cm².

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